

Course Code	AAI101/AAI201
Title of the Course	Introduction to AI and Prompt Engineering
Course Category	Engineering Science Course (ESC)
Credits	02
NCrF Hours	30
Tools Used	ChatGPT / Claude / Gemini / Copilot (free tier) — browser based

Main Focus of the Course Outcomes

Employability	Skill Development	Entrepreneurship
✓	✓	✓

Course Objectives:

1. Understand what Artificial Intelligence is, its history, and how it is changing every engineering field
2. Understand key AI concepts — Machine Learning, Deep Learning, and Neural Networks — at a conceptual level
3. Explore how AI is applied across domains — healthcare, agriculture, manufacturing, finance, and education
4. Understand Generative AI — how large language models (LLMs) like ChatGPT and Gemini work
5. Learn and practise Prompt Engineering — the skill of communicating effectively with AI tools to get useful results

Course Content

Unit	Topics	Weightage
1.	Introduction to Artificial Intelligence 1. Definition and scope of AI; brief history — from Alan Turing to ChatGPT	20%

	1. Types of AI — Narrow AI, General AI, Super AI (concept) 2. AI vs Machine Learning vs Deep Learning — what is the difference 3. How machines learn — data, patterns, and predictions (concept, no maths) 4. AI today — where it already exists in daily life (Google Maps, Spotify, Netflix, Google Photos, voice assistants) 5. Machine Learning Concepts, Learning Strategies- Supervised, Unsupervised, Reinforcement Learning 6. Data, Data Visualization, Data Pre-processing Class Activity: Students list 5 AI tools they use every day and discuss how each one 'learns'. <i>Branch connect: CE — AI in structural health monitoring; ME — AI in predictive maintenance; EE — smart grid; CSE — AI in compilers; BT — drug discovery; AG — crop yield prediction.</i>	
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Definition and Scope of Artificial Intelligence

Artificial Intelligence (AI) is a branch of computer science that deals with creating machines and computer systems that can perform tasks which normally require **human intelligence**.

These tasks include:

- Learning
- Reasoning
- Problem-solving
- Decision-making
- Understanding language
- Recognizing images and objects
- Understanding speech
- Planning

Artificial Intelligence is the ability of a machine or computer system to perform tasks that normally require human intelligence.

Scope of Artificial Intelligence

The **scope of AI** refers to the areas where AI can be used to perform intelligent tasks.

Major areas of AI:

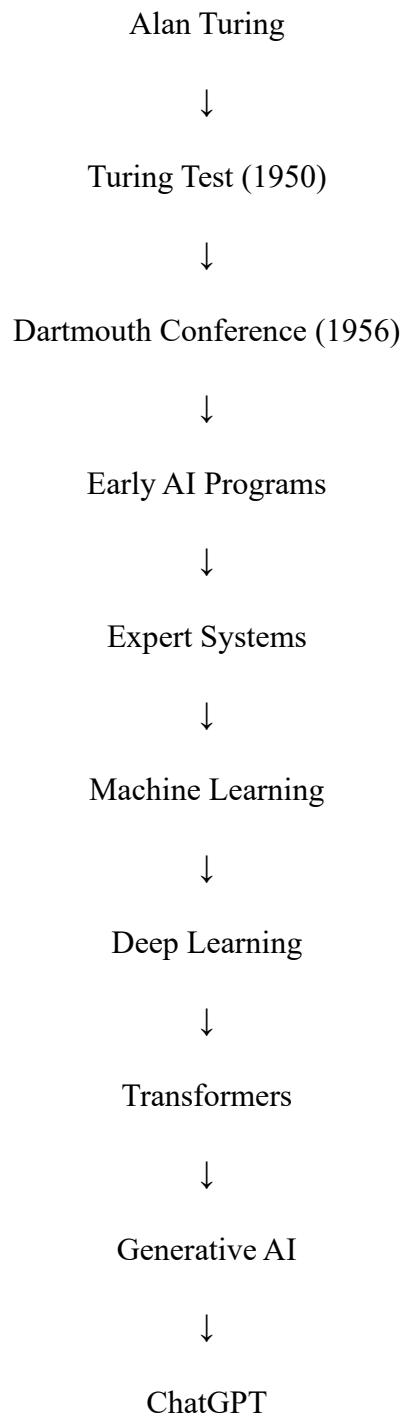
1. **Machine Learning** – Machines learn from data and make predictions.
2. **Deep Learning** – Uses neural networks to learn complex patterns.
3. **Natural Language Processing (NLP)** – Helps computers understand and generate human language.
4. **Computer Vision** – Helps machines understand images and videos.
5. **Robotics** – Uses AI to make robots perform intelligent tasks.
6. **Expert Systems** – Provides solutions or decisions using stored knowledge and rules.
7. **Speech Recognition** – Converts human speech into understandable information.
8. **Generative AI** – Generates new text, images, audio, video, or code.
9. **Healthcare** – Disease detection, medical image analysis, and drug discovery.
10. **Education** – AI tutors, personalized learning, and automated assessment.
11. **Finance** – Fraud detection, risk analysis, and financial prediction.
12. **Transportation** – Route planning, traffic prediction, and autonomous vehicles.

Brief History of AI — From Alan Turing to ChatGPT

Year/Period	Event	Importance
1950	Alan Turing proposed the Turing Test	Proposed a way to evaluate whether a machine could demonstrate human-like intelligent behavior.
1956	Dartmouth Conference	The term “ Artificial Intelligence ” was formally introduced as a field of study.
1950s–1960s	Early AI programs	Researchers developed programs capable of solving mathematical and logical problems.

Year/Period	Event	Importance
1960s–1970s	Early Natural Language and Robotics	AI research expanded into language understanding and robotics.
1970s–1980s	Expert Systems	AI systems were developed to solve specialized problems using expert knowledge and rules.
1980s–1990s	Machine Learning developed further	Researchers increasingly focused on allowing computers to learn from data.
1997	IBM Deep Blue defeated Garry Kasparov	Demonstrated the ability of computers to perform complex strategic tasks such as chess.
2012	Deep Learning breakthrough	Deep neural networks achieved major improvements in image recognition.
2017	Transformer architecture introduced	Transformers became an important foundation for modern language AI.
2020	GPT-3	Large language models demonstrated powerful text-generation capabilities.
2022	ChatGPT launched	Generative AI became widely accessible for conversation, writing, explanation, coding, and other tasks.
Today	Generative AI and modern AI	AI is widely used in education, healthcare, finance, transportation, entertainment, software development, and many other fields.

AI Development Flow



Types of Artificial Intelligence (AI)

Artificial Intelligence can be classified into three types based on its **capabilities and level of intelligence**:

1. Narrow AI (Weak AI)

Narrow AI is an AI system designed to perform a **specific task or a limited set of tasks**.

- It is designed for a particular purpose.
- It cannot perform tasks outside its designed area.
- Most AI systems used today are Narrow AI.
- It can perform its specific task very efficiently.

Examples:

- Google Maps for route planning
- Face recognition
- Spam email detection
- Voice assistants
- Recommendation systems of Netflix and Spotify

Example:

A chess-playing AI can play chess but cannot teach a mathematics class.

Narrow AI = AI designed for a specific task.

2. General AI (Strong AI)

General AI refers to an AI system that would have the ability to **understand, learn, and perform a wide range of intellectual tasks like a human being.**

- It would be capable of learning different types of tasks.
- It would not be limited to one specific task.
- It could apply knowledge from one area to another.
- It is a **theoretical concept** and has not been achieved in practice.

Example:

A hypothetical General AI could:

- Learn mathematics.
- Write programs.
- Understand languages.

- Play games.
- Solve problems.
- Learn new skills.

General AI = Human-like intelligence across many different tasks.

3. Super AI (Superintelligence)

Super AI refers to a hypothetical AI that would have **intelligence exceeding human intelligence** in virtually all areas.

- It would be more capable than humans in many intellectual tasks.
- It could potentially solve extremely complex problems.
- It is a theoretical concept.
- Super AI does not currently exist.

Hypothetical Examples:

A Super AI could potentially:

- Solve highly complex scientific problems.
- Learn extremely quickly.
- Make advanced decisions.
- Develop solutions beyond current human capabilities.

Super AI = Intelligence that exceeds human intelligence.

Difference Between Narrow AI, General AI and Super AI

Point	Narrow AI	General AI	Super AI
1. Meaning	Designed for a specific task.	Can perform a wide range of tasks like a human.	Would exceed human intelligence.
2. Scope	Limited	Broad	Beyond human-level

Point	Narrow AI	General AI	Super AI
3. Intelligence	Specialized intelligence	Human-like general intelligence	Greater than human intelligence
4. Tasks	Specific tasks	Multiple types of tasks	Potentially almost all intellectual tasks
5. Learning	Learns within a specific area	Could learn different areas like humans	Could potentially learn at a level beyond humans
6. Flexibility	Low	High	Very high
7. Human-Level Ability	No	Yes, theoretically	Exceeds humans, theoretically
8. Present Status	Exists today	Does not currently exist	Does not currently exist
9. Example	Face recognition, Google Maps	Hypothetical human-like AI	Hypothetical superintelligent AI
10. Main Idea	Specific intelligence	General intelligence	Superhuman intelligence

Functionality-Based Classification of AI

Functionality-based classification classifies Artificial Intelligence systems according to **how they function and interact with humans and their environment**.

There are **four types**:

1. Reactive Machines

These AI systems **only respond to the current situation**. They do not store past experiences or use memory to make future decisions.

Example: IBM Deep Blue chess-playing system.

No memory → Only current situation

2. Limited Memory

These AI systems can use **past information for a limited period** to make decisions.

Most modern AI applications fall into this category.

Examples:

- Self-driving cars
- Recommendation systems
- Chatbots using conversation context

Uses limited past information → Makes decisions

3. Theory of Mind

This refers to a **hypothetical AI** that would be able to understand human **emotions, beliefs, intentions, and thoughts**.

Such AI would be able to interact with humans in a more human-like way.

Example: No fully developed real-world example currently exists.

Understands human emotions and intentions

4. Self-Aware AI

This refers to a **hypothetical AI** that would have its own **consciousness, awareness, thoughts, and feelings**.

It would understand its own existence.

Example: No such AI currently exists.

Has self-consciousness and self-awareness

Type	Main Feature	Example/Status
Reactive Machines	Responds only to current situations	IBM Deep Blue
Limited Memory	Uses limited past information	Self-driving systems
Theory of Mind	Understands emotions and intentions	Hypothetical

Type	Main Feature	Example/Status
Self-Aware AI	Has consciousness and self-awareness	Hypothetical

Functionality-based classification of AI categorizes AI systems according to their ability to use memory, respond to the environment, understand humans, and possess self-awareness.

AI vs Machine Learning vs Deep Learning

Artificial Intelligence (AI), Machine Learning (ML), and Deep Learning (DL) are related concepts, but they are not the same. AI is the broadest concept, ML is a part of AI, and DL is a part of ML.

Point	Artificial Intelligence (AI)	Machine Learning (ML)	Deep Learning (DL)
1. Meaning	AI means making machines perform tasks that normally require human intelligence.	ML means making computers learn from data and improve automatically.	DL is a type of ML that uses deep neural networks to learn from data.
2. Relationship	AI is the largest field.	ML is a subset of AI.	DL is a subset of ML.
3. Main Aim	To make machines intelligent.	To make machines learn from data.	To make machines learn complex patterns automatically.
4. Working	AI can work using rules, logic, algorithms, or learning methods.	ML finds patterns in data and uses them to make predictions.	DL uses multiple layers of neural networks to find patterns.
5. Data Requirement	AI may work with or without large amounts of data.	ML generally needs data for learning.	DL usually needs a large amount of data.

Point	Artificial Intelligence (AI)	Machine Learning (ML)	Deep Learning (DL)
6. Human Involvement	Humans may provide rules and instructions to the system.	Humans usually select and prepare important features from data.	The system can automatically learn many useful features from raw data.
7. Feature Selection	Features may be manually defined depending on the AI method.	Feature selection is often done by humans.	Features are generally learned automatically by the neural network.
8. Algorithms Used	Uses rules, search algorithms, ML, DL, expert systems, etc.	Uses decision trees, regression, SVM, k-NN, etc.	Uses CNN, RNN, LSTM, Transformers, and other neural networks.
9. Computing Power	Depends on the type of AI system.	Usually requires less computing power than DL.	Usually requires high computing power, especially for training.
10. Complexity	Can be simple or highly complex.	Generally more complex than simple rule-based systems.	Usually highly complex because it contains many neural-network layers.
11. Learning Ability	AI systems may or may not learn from experience.	Learning from data is the main feature of ML.	Learns complex features and patterns through multiple layers.
12. Examples	Chatbots, robots, virtual assistants, game-playing systems.	Spam detection, recommendation systems, house-price prediction.	Face recognition, speech recognition, image recognition, generative AI.

Point	Artificial Intelligence (AI)	Machine Learning (ML)	Deep Learning (DL)
13. Applications	Robotics, automation, games, virtual assistants, decision-making.	Fraud detection, prediction, classification, recommendations.	Computer vision, speech recognition, natural language processing.
14. Neural Networks	Neural networks may or may not be used.	Neural networks can be used, but are not required.	Deep neural networks are the main technology used.
15. Example in Daily Life	A smart robot that can understand and respond to humans.	YouTube predicting videos you may like based on your previous activity.	Face unlock on a smartphone recognizing your face from an image.

Example

Suppose we want a computer to recognize cats and dogs:

- AI: The overall goal is to make the computer intelligent enough to recognize animals.
- ML: The computer learns from many examples of cats and dogs.
- DL: A deep neural network automatically learns features such as ears, eyes, shapes, and other patterns to recognize them.

AI is the broad field of making machines intelligent.

ML is a part of AI that allows machines to learn from data.

DL is a part of ML that uses deep neural networks to learn complex patterns automatically.

Natural Language Processing (NLP)

Natural Language Processing (NLP) is a branch of **Artificial Intelligence (AI)** that enables computers to **understand, process, interpret, and generate human language**.

Simple Definition

NLP is the technology that allows computers to communicate with and understand human language in the form of text or speech.

Examples of NLP

1. **Chatbots** – Understand and respond to user questions.
2. **Google Translate** – Translates text from one language to another.
3. **Voice Assistants** – Understand spoken commands.
4. **Spam Detection** – Identifies unwanted emails.
5. **Sentiment Analysis** – Identifies whether text expresses positive or negative feelings.
6. **Text Summarization** – Creates a shorter version of a long text.
7. **Speech-to-Text** – Converts spoken words into written text.

Applications of NLP

- Machine translation
- Chatbots and virtual assistants
- Speech recognition
- Text classification
- Sentiment analysis
- Information retrieval
- Text summarization

How Machines Learn — Data, Patterns, and Predictions

DATA => Find Patterns => LEARN => Create Model => New Data => PREDICTION

Machine learning allows a computer to learn from data and make predictions or decisions without being given every rule manually.

1. Data

Data is the information given to a machine for learning.

Examples:

- Pictures of cats and dogs
- Student marks and attendance
- Previous house prices

- Emails marked as spam or not spam

2. Patterns

A pattern is something that repeatedly appears in the data.

For example, if we give a machine many pictures of cats and dogs:

- Cats may have certain shapes, ears, eyes, etc.
- Dogs may have different shapes and features.

3. Learning

The machine studies the data and the patterns found in it.

It creates a model that represents what it has learned.

For example:

Data → Find Patterns → Learn → Create Model

The model can then be used on new data.

4. Prediction

After learning, the machine receives new data that it has not seen before.

It uses the patterns it learned to make a prediction.

Example:

If a machine has learned from thousands of cat and dog images, we give it a new image.

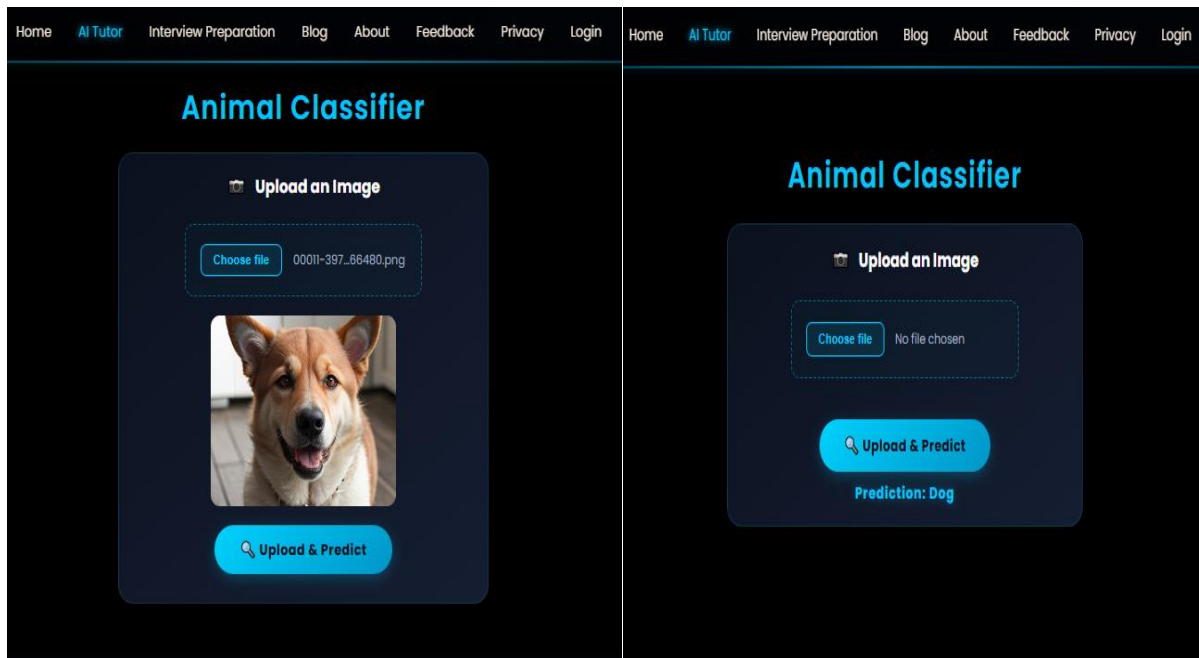
It may predict:

"This is a Cat."

5. Example

Suppose we want to predict whether a student will pass or fail.

Step	What happens
Data	Previous students' attendance, study hours, and marks are given.
Patterns	Machine finds that students with good attendance and regular study often pass.
Learning	Machine learns these patterns.
New Data	Information about a new student is given.
Prediction	Machine predicts whether the student is likely to pass or fail.



AI Today — Where AI Exists in Daily Life

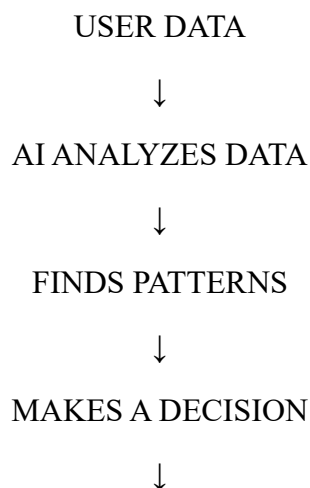
Artificial Intelligence (AI) is not only used in robots or advanced machines. **AI is already a part of our everyday life.** Many apps and services use AI to understand our behavior, recognize information, make recommendations, and help us make decisions.

No.	Daily-Life Example	How AI Is Used	Simple Example
1	Google Maps	AI analyzes traffic, routes, and travel information.	It suggests the fastest route to your destination.
2	Spotify	AI studies your listening habits and recommends music.	It suggests songs similar to the songs you usually listen to.
3	Netflix	AI analyzes your watching history and preferences.	It recommends movies and shows you may like.
4	Google Photos	AI recognizes faces, objects, places, and images.	It can group photos of the same person automatically.
5	Voice Assistants	AI understands human speech and responds to commands.	Siri, Google Assistant, and Alexa can answer questions or perform tasks.

No.	Daily-Life Example	How AI Is Used	Simple Example
6	YouTube	AI recommends videos based on your viewing behavior.	Your homepage shows videos related to your interests.
7	Email	AI identifies unwanted or suspicious emails.	Gmail automatically moves many spam emails into the Spam folder.
8	Online Shopping	AI studies your searches and purchases.	Amazon or other shopping apps recommend products you may want to buy.
9	Face Unlock	AI can recognize a person's face.	A smartphone unlocks when it recognizes the owner's face.
10	Social Media	AI recommends posts, reels, and content based on user activity.	Instagram shows content similar to what you frequently watch or like.
11	Online Banking	AI can help detect unusual or suspicious transactions.	A bank may flag a transaction that looks different from your normal activity.
12	Chatbots	AI understands questions and generates responses.	Customer-support chatbots answer common customer questions.

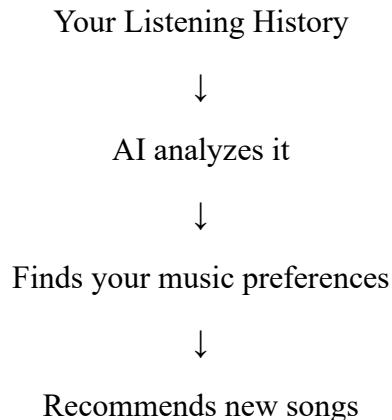
How AI Works in These Applications

Most of these applications follow a simple process:



GIVES RESULT / RECOMMENDATION

Example: Spotify



Machine Learning Concepts and Learning Strategies

Machine Learning (ML) is a part of Artificial Intelligence that allows computers to **learn from data and make predictions or decisions without being explicitly programmed for every situation.**

There are three main learning strategies:

1. **Supervised Learning**
2. **Unsupervised Learning**
3. **Reinforcement Learning**

1. Supervised Learning

In **Supervised Learning**, the machine learns from **labeled data**.

Here, we provide the machine with:

- **Input data**
- **Correct output**

The machine learns the relationship between them and then predicts the output for new data.

Simple Example

Suppose we want a computer to identify whether an email is **Spam or Not Spam**.

We give it many examples:

Email	Correct Answer
"You won a lottery!"	Spam
"Get free money now!"	Spam

Email	Correct Answer
"Meeting at 10 AM"	Not Spam
"Your assignment is ready"	Not Spam

The machine learns from these examples.

When a new email arrives, it predicts whether it is **Spam or Not Spam**.

Common Applications

- Spam email detection
- House price prediction
- Student result prediction
- Disease classification

2. Unsupervised Learning

In **Unsupervised Learning**, the machine learns from **unlabeled data**.

We give the machine data, but **we do not provide the correct answers**.

The machine tries to find:

- Groups
- Similarities
- Patterns
- Relationships

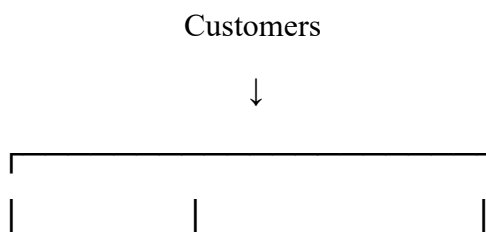
by itself.

Example

Suppose a shopping website has information about thousands of customers.

We don't tell the machine which customers belong to which group.

The machine may find groups such as:



Group 1	Group 2	Group 3
Frequent	Occasional	New
Buyers	Buyers	Customers

The groups are created based on similarities in customer behavior.

Common Applications

- Customer grouping
- Market segmentation
- Finding similar documents
- Recommendation systems
- Detecting unusual patterns

Cluster in Unsupervised Learning

A cluster is a group of data points that are more similar to each other than to data points belonging to other groups.

Clustering is a common **unsupervised learning technique** used to discover hidden groups or patterns in unlabeled data.

In **unsupervised learning**, the data does not have predefined labels. The algorithm examines the data and groups similar data points into **clusters** based on their characteristics or features.

Example

Suppose a shopping website has information about customers:

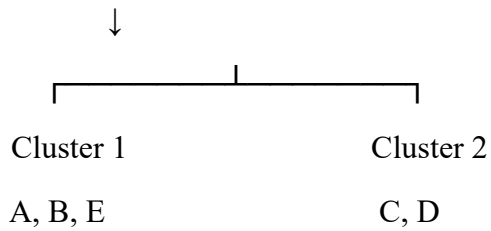
Customer	Monthly Purchases	Monthly Spending
A	10	₹20,000
B	12	₹22,000
C	2	₹3,000
D	3	₹4,000
E	8	₹15,000

An unsupervised learning algorithm may form:

Customers



Find Similarities



Here:

- **Cluster 1:** Customers with relatively high purchases and spending.
- **Cluster 2:** Customers with relatively low purchases and spending.

The algorithm creates these groups **without being told in advance what the groups should be called.**

Example algorithms: K-Means Clustering, Hierarchical Clustering.

3. Reinforcement Learning

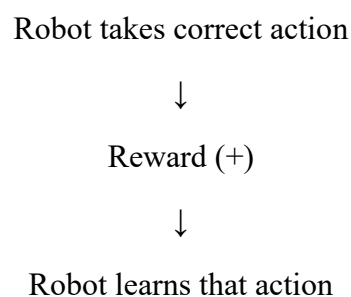
In **Reinforcement Learning**, a machine or **agent learns by interacting with its environment.**

The machine:

1. Takes an action.
2. Gets a **reward or penalty.**
3. Learns from the result.
4. Tries to improve its future actions.

Simple Example

Imagine teaching a robot to reach a destination.



If it takes a wrong action:

Robot takes wrong action

↓
Penalty (-)

↓
Robot tries another action

After many attempts, the robot learns which actions give better results.

Common Applications

- Game playing
- Robotics
- Self-driving systems
- Resource management
- Recommendation and decision systems

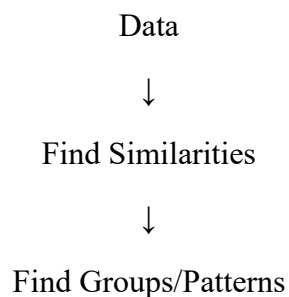
Difference Between Supervised, Unsupervised and Reinforcement Learning

Point	Supervised Learning	Unsupervised Learning	Reinforcement Learning
1.Learning Type	Learns from labeled data.	Learns from unlabeled data.	Learns through interaction.
2.Correct Answer	Correct answer is provided during training.	Correct answer is not provided.	No fixed answer is provided; rewards guide learning.
3.Main Purpose	Make predictions or classifications.	Find hidden patterns or groups.	Find the best actions to achieve a goal.
4. Feedback	Gets feedback through known correct outputs.	Usually no direct feedback.	Gets rewards or penalties.
5. Data	Labeled data.	Unlabeled data.	Experience from actions and outcomes.
6.Learning Method	Learns input-output relationships.	Finds similarities and patterns.	Learns which actions produce better results.
7. Example	Spam detection.	Customer grouping.	Teaching a robot to move.

Point	Supervised Learning	Unsupervised Learning	Reinforcement Learning
8. Output	Known categories or predicted values.	Groups, patterns, or relationships.	Actions or decisions.
9. Human Role	Humans provide labeled training data.	Humans provide data but generally not labels.	Humans define the environment, actions, and reward system.
10.Common Algorithms	Linear Regression, Decision Tree, SVM, etc.	K-Means, Hierarchical Clustering, etc.	Q-Learning, Deep Q-Networks, etc.

Unsupervised Learning

Student gets data but no answers



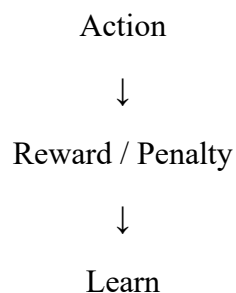
Example:

Machine receives many animal pictures and tries to group similar pictures together without being told their names.

Reinforcement Learning

Reinforcement learning is a learning method in which an agent learns by interacting with an environment and receiving rewards or penalties for its actions.

Learn by trial and error





Better Action

Example:

A computer learns to play a game by trying different moves and receiving rewards for successful moves.

2. Passive vs Active Reinforcement Learning

The main difference is **whether the agent decides which action to take**.

Point	Passive Reinforcement Learning	Active Reinforcement Learning
1. Meaning	Agent follows a fixed policy .	Agent chooses actions and learns the best policy.
2. Decision Making	Does not decide which action to take.	Decides which action to take.
3. Main Goal	Learn how good the states/actions are under a given policy.	Learn which actions are best to achieve the goal.
4. Policy	Policy is already given.	Policy is learned by the agent.
5. Exploration	Little or no exploration is required.	Exploration is important.
6. Learning	Learns from the results of following the fixed policy.	Learns by trying different actions and observing their results.
7. Example	Robot is told to always follow a particular route and learns which states are good or bad.	Robot chooses different routes to find the best route.
8. Action Selection	Actions are determined by the given policy.	Actions are selected by the agent.
9. Flexibility	Less flexible.	More flexible.
10. Simple Idea	“Follow and learn.”	“Try, learn, and choose.”

Example

Imagine a robot learning to reach a room.

Passive RL:

The robot is given a route → follows the route → learns which situations give good or bad results.

Active RL:

The robot can choose different routes → tries them → receives rewards/penalties → learns the best route.

Passive = Policy is given

Active = Policy is learned

Is Reinforcement Learning an Appropriate Abstract Model for Human Learning?

Yes, reinforcement learning can be considered a useful abstract model for some aspects of human learning, but it is not a complete model of human learning.

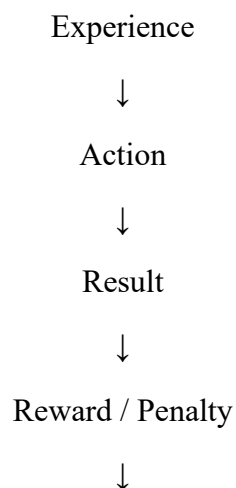
Why is it useful?

Humans often learn through **experience, rewards, and consequences**.

For example:

- A student studies hard and gets good marks → **positive reward**
- A person makes a mistake and faces a negative result → **learns to avoid that action**
- A child receives praise for good behavior → **more likely to repeat it**

This is similar to reinforcement learning:



Learning



Better Future Action

But Human Learning Is More Complex

Human beings do not learn only through rewards and penalties. Humans can also learn through:

- Reading and studying
- Observation
- Imitation
- Language
- Reasoning
- Memory
- Communication with others
- Understanding concepts

Therefore:

Reinforcement learning is a useful simplified model for understanding learning from experience and feedback, but it cannot explain all types of human learning.

Data, Data Visualization and Data Pre-processing

These are important concepts in **Machine Learning** because a machine learns from data. Before giving data to a machine-learning model, it is often necessary to understand, clean, and prepare the data.

1. Data

Data is a collection of information that can be used for analysis and decision-making.

Data can be in the form of:

- Numbers
- Text
- Images
- Audio
- Video
- Tables

Example

Student data:

Name	Age	Attendance	Marks
Rahul	18	85%	78
Priya	19	92%	88
Aman	18	70%	65

Here, **Name**, **Age**, **Attendance**, and **Marks** are examples of data.

Types of Data

Type	Meaning	Example
Structured Data	Data arranged in a fixed format such as rows and columns.	Student table
Unstructured Data	Data without a fixed structure.	Images, videos, audio
Semi-structured Data	Data that has some organization but not a fixed table format.	JSON, XML

In Machine Learning

Data is used to:

1. Train the machine-learning model.
2. Test the model.
3. Make predictions on new information.

2. Data Visualization

Data Visualization is the process of representing data graphically using graphs, charts, or diagrams so that it becomes easier to understand.

Instead of looking at a large table, we can use a graph to quickly understand the information.

Common Types

Visualization	Used For
Bar Chart	Comparing different categories
Line Chart	Showing changes over time

Visualization	Used For
Pie Chart	Showing parts of a whole
Histogram	Showing distribution of numerical data
Scatter Plot	Showing the relationship between two variables

Example

Suppose the marks of students are:

Rahul → 78

Priya → 88

Aman → 65

Neha → 92

A **bar chart** can be used to easily compare their marks.

Why Data Visualization is Used?

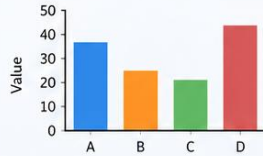
1. Makes data easier to understand.
2. Helps identify patterns.
3. Helps identify trends.
4. Helps find unusual values.
5. Makes comparison easier.
6. Helps in making better decisions.

Types of Charts in Data Visualization

(with simple examples)

1. Bar Chart

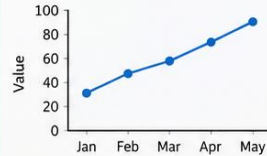
Used to compare different categories.



Example: Sales of different products

2. Line Chart

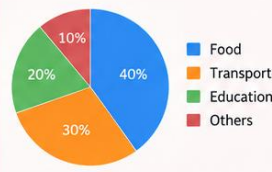
Used to show changes over time.



Example: Monthly temperature

3. Pie Chart

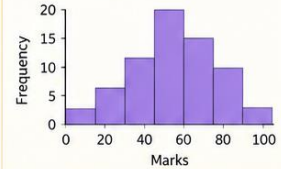
Used to show parts of a whole.



Example: Monthly expenses

4. Histogram

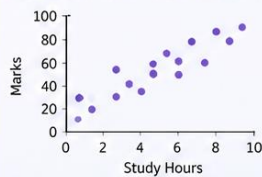
Used to show the distribution of numerical data.



Example: Distribution of student marks

5. Scatter Plot

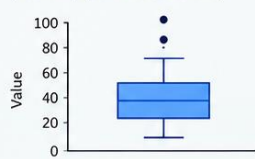
Used to show the relationship between two numerical variables.



Example: Study hours vs Marks

6. Box Plot

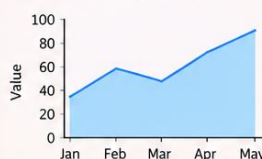
Used to show the spread of data (median, quartiles, outliers).



Example: Marks of students

7. Area Chart

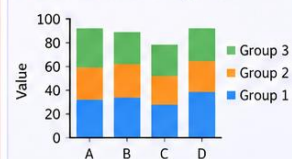
Used to show change in quantity over time with filled area.



Example: Website visitors over months

8. Stacked Bar Chart

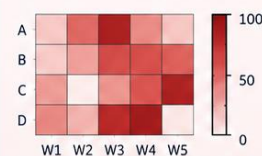
Used to show parts of a whole across different categories.



Example: Sales by category and region

9. Heatmap

Used to show data values using colors (good for large data).



Example: Performance of students across weeks

10. Treemap

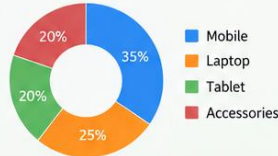
Used to show hierarchical data using nested rectangles.



Example: Product categories in a store

11. Donut Chart

Similar to pie chart but with a hole in the center.



Example: Device usage

12. Radar Chart

Used to compare multiple variables.



Example: Student performance in subjects

Different charts are used for different types of data and purposes.
Choose the right chart to make your data easy to understand!

3. Data Pre-processing

Data Pre-processing is the process of **cleaning and preparing raw data before using it for Machine Learning**.

Raw data may contain:

- Missing values
- Incorrect values
- Duplicate data
- Different formats
- Unwanted information

Therefore, data needs to be prepared before giving it to the ML model.

Example

Suppose we have:

Student	Age	Marks
Rahul	18	78
Priya	Missing	88
Aman	18	-5
Rahul	18	78

Here:

- Age of Priya is missing.
- Aman has an incorrect marks value.
- Rahul's record is duplicated.

These problems should be handled before training the model.

Main Steps of Data Pre-processing

1. Data Collection

Collect the required data from different sources.

Example:

Student information collected from forms, databases, or files.

2. Data Cleaning

Remove or correct problems present in the raw data.

It includes:

- **Handling Missing Values: -**

Missing values can be:

- Removed
- Replaced with a suitable value
- Filled using other methods

Example:

Age = 18, 19, Missing, 20

The missing value can be filled appropriately.

- **Data Scaling: -**

Bringing numerical values into a suitable range when required.

Example:

Age: 18, 20, 22

Salary: 25000, 50000, 75000

These values have very different scales, so scaling may be useful for some ML algorithms.

- **Removing duplicate records: -**

Repeated records are identified and removed when they are not needed.

- **Handling Outliers: -**

An **outlier** is a value that is very different from most other values.

Example:

10, 12, 11, 13, 12, 150

Here, **150** may be an outlier depending on the context.

- Correcting incorrect data
- Handling inconsistent data
- Handling outliers

Example:

Age: 18, 19, Missing, 20

The missing value needs to be handled before using the data.

3. Data Integration

Combine data collected from different sources into a single dataset.

Example:

Student Database + Attendance Database

↓

Combined Dataset

4. Data Transformation

Convert data into a suitable format for analysis or machine learning.

It includes:

- Normalization
- Standardization

- Encoding categorical data
- Converting data into suitable formats

Example:

Gender:

Male → 0

Female → 1

5. Data Reduction

Reduce the size or complexity of data while keeping the important information.

It can include:

- Removing unnecessary features
- Selecting important data
- Reducing the number of records when appropriate

Example:

If a dataset has 50 columns but only 10 are useful for prediction, unnecessary columns can be removed.

6. Data Splitting

Divide the prepared data into different parts for machine learning.

Commonly:

- Training data: Used to train the model.
- Testing data: Used to check the model's performance.

Main Step	What is done?
1. Data Collection	Collect the required data.
2. Data Cleaning	Remove or correct errors, missing values, duplicates, etc.
3. Data Integration	Combine data from different sources.
4. Data Transformation	Convert data into a suitable format.
5. Data Reduction	Remove unnecessary data and reduce complexity.
6. Data Splitting	Divide data into training and testing sets.

Difference Between Data, Data Visualization and Data Pre-processing

Point	Data	Data Visualization	Data Pre-processing
1. Meaning	Collection of information.	Graphical representation of data.	Process of preparing data for use.
2. Purpose	Provides information for analysis.	Makes information easier to understand.	Makes data suitable for ML models.
3. Form	Numbers, text, images, etc.	Charts, graphs, and diagrams.	Cleaned and transformed data.
4. Example	Student marks.	Bar chart of student marks.	Removing missing or incorrect marks.
5. Main Activity	Collecting information.	Representing information visually.	Cleaning and transforming information.
6. Used Before ML?	Yes.	Helps understand the data.	Yes, commonly before training an ML model.

Unit - 2

Machine Learning: Classification Technique — K-Nearest Neighbors (KNN)

1. What is Classification?

Classification is a supervised machine learning technique used to **divide data into different predefined categories or classes**.

Examples:

- Email → **Spam / Not Spam**
- Student → **Pass / Fail**
- Disease → **Yes / No**
- Image → **Cat / Dog**

2. What is K-Nearest Neighbors (KNN)?

K-Nearest Neighbors (KNN) is a simple **supervised machine learning classification algorithm**.

It classifies a new data point by looking at the **K nearest data points** that are already known.

Simple idea:

A new data point usually belongs to the class of its nearest neighbors.

KNN does not build a complicated model during training. Instead, it stores the existing data and compares a new data point with it when a prediction is required.

3. How KNN Works

Suppose we have two classes:

-  Class A
-  Class B

For a new point, KNN follows these steps:

Step 1: Choose K

Select the number of neighbors, such as: **K = 3**

Step 2: Calculate Distance

Calculate the distance between the new point and existing data points.

The commonly used distance is **Euclidean Distance**.

Step 3: Find K Nearest Points

Select the **K points having the smallest distances**.

Step 4: Voting

Check the classes of those K nearest points.

Step 5: Assign the Class

The class having the **maximum number of votes** becomes the class of the new data point.

4. KNN Example

Suppose we want to classify a student based on:

- **Study Hours**
- **Marks**

We have the following data:

Student	Study Hours (X)	Marks (Y)	Class
A	2	50	Fail
B	3	60	Fail
C	4	70	Pass
D	5	80	Pass
E	6	90	Pass

New student:

P = (4, 75)

Let: **K = 3**

We calculate the distance of P from every student.

5. Euclidean Distance

The formula for two points:

$$\begin{array}{l} (x_1, y_1) \\ \text{and} \\ (x_2, y_2) \\ \text{is:} \\ d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \end{array}$$

Find the difference between the X values and Y values, square them, add them, and take the square root.

6. Numerical Problem

Given:

New point:

$$P = (4, 75)$$

Existing points:

- $A = (2, 50) \rightarrow \text{Fail}$
- $B = (3, 60) \rightarrow \text{Fail}$
- $C = (4, 70) \rightarrow \text{Pass}$
- $D = (5, 80) \rightarrow \text{Pass}$
- $E = (6, 90) \rightarrow \text{Pass}$

We need to find the **3 nearest neighbors**.

Distance from A

$$\begin{aligned}d_A &= \sqrt{(4 - 2)^2 + (75 - 50)^2} \\&= \sqrt{2^2 + 25^2} \\&= \sqrt{4 + 625} \\&= \sqrt{629} \\d_A &\approx 25.08\end{aligned}$$

Distance from B

$$\begin{aligned}d_B &= \sqrt{(4 - 3)^2 + (75 - 60)^2} \\&= \sqrt{1^2 + 15^2} \\&= \sqrt{226} \\d_B &\approx 15.03\end{aligned}$$

Distance from C

$$\begin{aligned}d_C &= \sqrt{(4 - 4)^2 + (75 - 70)^2} \\&= \sqrt{0 + 25} \\d_C &= 5\end{aligned}$$

Distance from D

$$\begin{aligned}d_D &= \sqrt{(4 - 5)^2 + (75 - 80)^2} \\&= \sqrt{1 + 25} \\&= \sqrt{26} \\d_D &\approx 5.10\end{aligned}$$

Distance from E

$$\begin{aligned}d_E &= \sqrt{(4 - 6)^2 + (75 - 90)^2} \\&= \sqrt{4 + 225} \\&= \sqrt{229} \\d_E &\approx 15.13\end{aligned}$$

7. Arrange Distances

Student	Distance	Class
C	5.00	Pass
D	5.10	Pass
B	15.03	Fail
E	15.13	Pass
A	25.08	Fail

Since:

$$K = 3$$

we select the **3 nearest neighbors**:

1. $C \rightarrow \text{Pass}$
2. $D \rightarrow \text{Pass}$
3. $B \rightarrow \text{Fail}$

Voting

Class	Votes
Pass	2
Fail	1

Therefore:

New Student = Pass

8. KNN Algorithm

New Data Point



Select value of K



Calculate distances



Find K nearest points



Check their classes



Majority voting



Predicted Class

Example: Classifying a Student

Suppose we have students based on their **study hours** and **marks**.

Student	Study Hours	Marks	Result
A	2	40	Fail
B	3	50	Fail
C	4	70	Pass
D	5	80	Pass
E	6	90	Pass

Now we have a **new student X**:

Study Hours = 4.5

Marks = 75

We want to predict whether the student will **Pass or Fail**.

Step 1: Choose K

Suppose: **K = 3**

This means KNN will look at the **3 nearest students**.

Step 2: Find the nearest students

After comparing the new student with the existing students, suppose the 3 nearest students are:

Neighbor	Result
C	Pass
D	Pass
B	Fail

Step 3: Voting

Now count the results:

Pass → 2, Fail → 1 vote

Pass has more votes.

Therefore:

Prediction = PASS

Important Questions

- 1). Explain the historical background and evolution of artificial intelligence.
- 2). Provide a concise definition of artificial intelligence and its main objectives.
- 3). Discuss the role of ai in future of technology and society.
- 4). Explain the term "Artificial Intelligence". Also give the major characteristics of AI.
- 5). What do you mean by AI? Explain the use of AI in various fields.
- 6). Define Artificial Intelligence and mention any two applications of AI.
- 7). Write a short note on the foundations of artificial intelligence.
- 8). Discuss various application domains of machine learning.
- 9). Explain Intelligence and Artificial Intelligence. How do you distinguish between the two??
- 10). Explain any three-application area of AI in short.
- 11). Explain the historical background and evolution of AI, highlighting major milestones.
- 12). Define the term reinforcement learning. Explain how passive reinforcement learning differ from active reinforcement learning.
- 13). Explain in brief the concept of reinforcement learning.
- 14). Explain Supervised Learning and Unsupervised Learning with examples.
- 15). Differentiate supervised and unsupervised learning.
- 16). What is machine learning? Differentiate between supervised, unsupervised and reinforcement learning?
- 17). Define reinforcement learning. Difference between passive and active reinforcement learning. Is for evolution reinforcement learning an appropriate abstract model for human learning?
- 18). Explain Nearest Neighbor rule used for classification.
- 19). Describe the four categories under which AI is classified with examples.
- 20). List various criteria for success in AI.